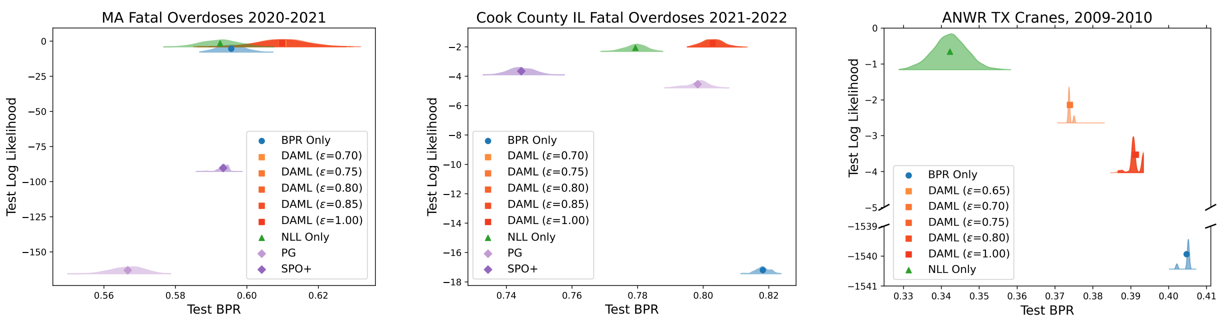


Figure 3: **Pareto frontier of best possible reach (BPR, x-axis) and log likelihood (LL, y-axis) for real-world tasks**. Higher is better on both axes. Each panel how the final models estimated by different training methods score on the test set of a forecasting task defined in Sec. 5. To capture the stochasticity of BPR due to our sampling-based ranking estimator, for each model we show an estimated density for BPR over 1000 trials. Across all 3 tasks, our proposed decision-aware ML (DAML) delivers better top-K decisions as measured by BPR than ML estimation. DAML also delivers likelihood comparable to ML methods and *much better* than directly optimizing BPR. Additionally, we compare to other decision-aware losses: Perturbed Gradient (PG) and Smart "Predict, then Optimize" (SPO+), both in the PyEPO package. For these methods, we use our ratio estimator to rank sites and the gradient of this estimator in Eq. (10). We conduct a hyperparameter search over learning rate (and perturbation noise for PG), and select the model with the best loss value on validation data. For overdose forecasting in MA (left) and Cook County IL (center), these decision-aware losses do indeed produce good top-K decisions, with SPO+ outperforming the ML estimate in Massachusetts and PG doing likewise in Cook County. However, as these models do not train to maximize likelihood, we accordingly get low-likelihood solutions. Furthermore, in terms of BPR they are outperformed by our DAML. On the ANWR TX Cranes task (right), unfortunately, both SPO+ and PG produced numerically unstable solutions - the parameters of the prediction model parameterize a Negative Binomial solution, and the parameters reached invalid values even with appropriate reparameterization. Despite a similar hyperparameter search, we were unable to effectively train SPO+ or PG during the 6-day response period.